

So far, we have seen that

$$|\int \xi dP| \leq \int |\xi| dP$$

our next question is whether following holds:

$$|E(\xi|\mathcal{G})| \leq E(|\xi||\mathcal{G}) \quad \text{a.s.}$$

The notion of convexity will play a key role to prove above inequality.

Definition (Convexity). A function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is said to be convex if for all $x, y \in \mathbb{R}$ and $\lambda \in [0, 1]$,

$$\varphi(\lambda x + (1 - \lambda)y) \leq \lambda \varphi(x) + (1 - \lambda)\varphi(y).$$

Theorem 2.2 (Jensen's inequality). Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be convex. If ξ and $\varphi(\xi)$ are integrable, then

$$\varphi(E(\xi|\mathcal{G})) \leq E(\varphi(\xi)|\mathcal{G}) \quad \text{a.s.}$$

Proof. Choose a supporting affine function such that

$$a_k x + b_k \leq \varphi(x)$$

at each rational point so that

$$\varphi(x) = \sup_{k \geq 1} (a_k x + b_k).$$

For each fixed $k \geq 1$, the inequality

$$a_k \xi + b_k \leq \varphi(\xi)$$

shows that

$$a_k E(\xi|\mathcal{G}) + b_k = E(a_k \xi + b_k|\mathcal{G}) \leq E(\varphi(\xi)|\mathcal{G}) \quad \text{a.s.}$$

There are countably many rational k , so these inequalities hold simultaneously on a set of probability one, which is equivalent to holding simultaneously outside one null set. Taking their supremum over k gives

$$\varphi(E(\xi|\mathcal{G})) = \sup_{k \geq 1} (a_k E(\xi|\mathcal{G}) + b_k) \leq E(\varphi(\xi)|\mathcal{G}) \quad \text{a.s.}$$

□

Remark.

1. Since $|\cdot|$ is convex, we have

$$|E(\xi|\mathcal{G})| \leq E(|\xi||\mathcal{G}) \quad \text{a.s.}$$

which we wanted to prove.

2. If $p \geq 1$ and $\xi \in L^p$, then

$$|E(\xi|\mathcal{G})|^p \leq E(|\xi|^p|\mathcal{G}) \quad \text{a.s.}$$

which implies that

$$E(|E(\xi|\mathcal{G})|^p) \leq E(|\xi|^p).$$

3. Martingales in Discrete Time

3.1 Sequences of Random Variables

From now on, we consider ξ_n as

$$\xi_n : \Omega \rightarrow \mathbb{R}, \quad n \in \mathbb{N}.$$

Definition. Let (ξ_n) be a sequence of random variables. For a fixed outcome $\omega \in \Omega$, the sequence

$$\xi_1(\omega), \xi_2(\omega), \dots$$

is called a **sample path** of (ξ_n) .

3.2 Filtrations

Definition. A **filtration** is a sequence of σ -fields $(\mathcal{F}_n)$ such that

$$\mathcal{F}_1 \subset \mathcal{F}_2 \subset \dots \subset \mathcal{F}.$$

Example 3.1. Let $\xi_n = 1$ for heads and $\xi_n = 0$ for tails on toss n , and put

$$\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n).$$

Let

$$A = \{\text{the first 5 tosses produce at least 2 heads}\}$$

At $n = 5$, we can decide whether A has occurred or not. But at $n = 4$, if the outcomes of the first four tosses are $TTHT$, then A remains undecided. Therefore,

$$A \in \mathcal{F}_5, \quad A \notin \mathcal{F}_4.$$

If the first four tosses are $THTH$, then it is possible to tell that A has occurred already at $n = 4$. However, it does not mean $A \in \mathcal{F}_4$.

Exercise 3.1. Find the smallest $n \geq 1$, if one exists, for which each of the following events belongs to $\mathcal{F}_n$.

1. $A = \{\text{the first occurrence of heads is preceded by no more than 10 tails}\}$
2. $B = \{\text{there is at least 1 head in the sequence } \xi_1, \xi_2, \dots\}$
3. $C = \{\text{the first 100 tosses produce the same outcome}\}$
4. $D = \{\text{there are no more than 2 heads and 2 tails among the first 5 tosses}\}$

Proof.

1. The event A occurs when at least one head appears among the first 11 tosses. Hence,

$$A = \bigcup_{k=1}^{11} \{\xi_k = 1\} \in \mathcal{F}_{11}.$$

After ten tails, a head on toss 11 makes A occur, whereas a tail makes it fail. Hence, $A \notin \mathcal{F}_{10}$ and the smallest n is 11.

2. After any finite string of n tails, a head may occur later or all subsequent tosses may be tails. Therefore, no finite n satisfies

$$B \in \mathcal{F}_n.$$

3. Note that

$$C = \{\xi_1 = \xi_2 = \cdots = \xi_{100} = 1\} \cup \{\xi_1 = \xi_2 = \cdots = \xi_{100} = -1\} \in \mathcal{F}_{100}.$$

After 99 heads, its occurrence still depends on toss 100. So $C \notin \mathcal{F}_{99}$.

4. $D = \emptyset \in \mathcal{F}_1$

□

Definition. A sequence (ξ_n) is **adapted to a filtration** $(\mathcal{F}_n)$ if

$$\xi_n \text{ is } \mathcal{F}_n\text{-measurable for every } n \geq 1.$$

Example 3.2. Every sequence (ξ_n) is adapted to its natural filtration $(\mathcal{F}_n)$ defined by

$$\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n).$$

Exercise 3.2. Show that

$$\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$$

is the smallest filtration to which (ξ_n) is adapted.

More precisely, if (ξ_n) is adapted to another filtration $(\mathcal{G}_n)$, then

$$\mathcal{F}_n \subset \mathcal{G}_n \quad \text{for all } n \in \mathbb{N}.$$

Proof. Since (ξ_n) is $\mathcal{G}_n$ -measurable, and

$$\mathcal{G}_1 \subset \mathcal{G}_2 \subset \cdots \subset \mathcal{G},$$

all of $\xi_1, \dots, \xi_n$ are $\mathcal{G}_n$ -measurable. Therefore,

$$\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n) \subset \mathcal{G}_n.$$

□